

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 January 2026
Team ID	LTVIP2026TMIDS86494
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion	Load electricity consumption data from CSV/Excel files
FR-2	Data Preprocessing	Support multiple datasets (year-wise / region-wise data)
FR-3	Data Analysis	Analyze electricity consumption by year (2019, 2020)
FR-4	Visualization	Create charts (bar, line, pie, map) for consumption analysis
FR-5	Interactive Dashboard	Provide filters for year, state, and region
FR-6	Comparative Analysis	Compare electricity consumption across different years
FR-7	Storytelling	Create a Tableau story for step-by-step insight explanation
FR-8	Reporting & Insights	Summarize key findings from dashboards

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to understand, visually clear, and user-friendly, allowing users to analyze electricity consumption with minimal effort.
NFR-2	Security	The data used for analysis should be stored securely, with controlled access to prevent unauthorized modification or misuse.
NFR-3	Reliability	The system should consistently display accurate electricity consumption data without errors or data loss.
NFR-4	Performance	Dashboards and visualizations should load quickly and respond smoothly to user interactions such as filters and selections.

NFR-5	Availability	The Tableau dashboards should be accessible whenever required, ensuring continuous availability for analysis and reporting.
NFR-6	Scalability	The system should handle additional datasets, more years of data, and expanded regional information without performance degradation.